

Complex Number

✓ The number having the form $x+iy$ is called complex number, where x and y are called real numbers and ' i ' which is imaginary unit has the property $i^2 = -1$ or $i = \sqrt{-1}$.

If $z = x+iy$ be a complex number then x is called the real part of z and is denoted by $\text{Re}\{z\}$ and y is called imaginary part of z and it is denoted by $\text{Im}\{z\}$.

Complex Conjugate: If $z = x+iy$ be a complex number, then the complex conjugate of z is denoted by $\bar{z}$ or z^* and is defined by

$$\bar{z} = \overline{x+iy} = x-iy.$$

Complex Variables (MID)

(a) ছোট খাতার প্রথম পৃষ্ঠা. (১০১ + ১০০) এর মোট মূল্য

(b) $\frac{1+2i}{1-3i}$ $\left(\frac{r e^{i\theta}}{r e^{i\phi}} \right) = \frac{r}{r} e^{i(\theta-\phi)}$

$$= \frac{(1+2i)(1+3i)}{(1-3i)(1+3i)} \quad \text{[work]}$$

$$= \frac{1+3i+2i+6i^2}{1+9i^2} = \frac{1+5i-6}{1-9} = \frac{-5+5i}{-8}$$

$$= \frac{5i-5}{10}$$

$$= \frac{5(i-1)}{10} = \frac{i-1}{2}$$

Polar form, $z = -\frac{1}{2} + \frac{1}{2}i$

Modulus, $r = \sqrt{x^2 + y^2}$

$$= \sqrt{\left(-\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^2}$$

$$= \sqrt{\frac{1}{4} + \frac{1}{4}} = \sqrt{\frac{2}{4}} = \sqrt{\frac{1}{2}}$$

$$= \sqrt{\frac{2}{4}} = \sqrt{\frac{1}{2}}$$

Argument, $\theta = \pi - \tan^{-1} \left| \frac{y}{x} \right|$

$$= \pi - \tan^{-1} \left| \frac{\frac{1}{2}}{-\frac{1}{2}} \right|$$

$$= \pi - \tan^{-1} |1| = \pi - \frac{\pi}{4} = \frac{3\pi}{4}$$

(2007) (2010) (2012) (2013) (2014) (2015) (2016) (2017) (2018) (2019) (2020) (2021) (2022) (2023) (2024) (2025) (2026) (2027) (2028) (2029) (2030) (2031) (2032) (2033) (2034) (2035) (2036) (2037) (2038) (2039) (2040) (2041) (2042) (2043) (2044) (2045) (2046) (2047) (2048) (2049) (2050) (2051) (2052) (2053) (2054) (2055) (2056) (2057) (2058) (2059) (2060) (2061) (2062) (2063) (2064) (2065) (2066) (2067) (2068) (2069) (2070) (2071) (2072) (2073) (2074) (2075) (2076) (2077) (2078) (2079) (2080) (2081) (2082) (2083) (2084) (2085) (2086) (2087) (2088) (2089) (2090) (2091) (2092) (2093) (2094) (2095) (2096) (2097) (2098) (2099) (2100)

Polar form, $Z = r(\cos \theta + i \sin \theta)$

$$= \frac{1}{\sqrt{2}} \left(\cos \frac{3\pi}{4} + i \sin \frac{3\pi}{4} \right) \quad \text{Ans}$$

Now,

$$\frac{2+3i}{3-7i} = \frac{(2+3i)(3+7i)}{(3-7i)(3+7i)}$$

$$= \frac{6+14i+9i+21i^2}{3^2-(7i)^2}$$

$$= \frac{6+14i+9i-21}{9+49}$$

$$= \frac{44i-15}{58}$$

$$= \frac{44i}{58} - \frac{15}{58}$$

$$= \frac{23}{58}i - \frac{15}{58} = -\frac{15}{58} + \frac{23}{58}i$$

$$r = \sqrt{x^2 + y^2}$$

$$= \sqrt{\left(-\frac{15}{58}\right)^2 + \left(\frac{23}{58}\right)^2}$$

$$= \frac{\sqrt{758}}{58}$$

$$\theta = \pi - \tan^{-1} \left(\left| \frac{y}{x} \right| \right) \text{ pro } = (\pi, \pi) \text{ pro } (b)$$

$$= \pi - \tan^{-1} \left| \frac{\frac{23}{58}}{-\frac{15}{58}} \right|, \text{ most value of } \theta$$

$$= \pi - \tan^{-1} \left| \frac{23}{15} \right|$$

Now,

$$Z = r(\cos \theta + i \sin \theta)$$

$$= \frac{\sqrt{758}}{58} \left(\cos \left(\pi - \tan^{-1} \left| \frac{23}{15} \right| \right) + i \sin \left(\pi - \tan^{-1} \left| \frac{23}{15} \right| \right) \right)$$

$$(x + iy)$$

Ans.

(C) Absolute Value: The absolute value of a complex number is the distance from the origin in the complex plane.

$$\text{if, } Z = x + iy$$

The absolute value is, pro = (x, y) pro

$$|Z| = \sqrt{x^2 + y^2}$$

$$(x, y) = (x, y) (7)$$

Argument: The argument of a complex number is the angle that the line joining the number to the origin makes with the positive real axis.

$$\text{for, } Z = x + iy$$

$$\text{Arg}(z) = \tan^{-1} \left(\frac{y}{x} \right)$$

$$(d) \arg(z_1 z_2) = \arg(z_1) + \arg(z_2)$$

Let in polar form,

$$z_1 = r_1 e^{i\theta_1}$$

$$\therefore \arg(z_1) = \theta_1$$

$$z_2 = r_2 e^{i\theta_2}$$

$$\therefore \arg(z_2) = \theta_2$$

Now,

$$z_1 z_2 = r_1 e^{i\theta_1} \cdot r_2 e^{i\theta_2}$$

$$= r_1 r_2 e^{i(\theta_1 + \theta_2)}$$

$$= r_1 r_2 e^{i(\theta_1 + \theta_2)}$$

$$\arg(z_1 z_2) = (\theta_1 + \theta_2)$$

$$\arg(z_1 z_2) = \arg(z_1) + \arg(z_2) \text{ (proved)}$$

$$(e) |z_1 z_2| = |z_1| |z_2|$$

In polar form,

$$\text{Let, } z_1 = r_1 e^{i\theta_1}$$

$$\therefore |z_1| = r_1$$

$$\text{and } z_2 = r_2 e^{i\theta_2}$$

$$\therefore |z_2| = r_2$$

$$\text{Now, } z_1 z_2 = r_1 e^{i\theta_1} \cdot r_2 e^{i\theta_2}$$

$$= r_1 r_2 e^{i(\theta_1 + \theta_2)}$$

$$|z_1 z_2| = r_1 r_2 \therefore |z_1 z_2| = |z_1| |z_2| \text{ (proved)}$$

(f) Given,

$$|z+3| + |z-3| = 10$$

Let $z = x+iy$

$$\Rightarrow |x+iy+3| + |x+iy-3| = 10$$

$$\Rightarrow |x+3+iy| + |x-3+iy| = 10$$

$$\Rightarrow \sqrt{(x+3)^2 + (y)^2} + \sqrt{(x-3)^2 + (y)^2} = 10$$

$$\Rightarrow \sqrt{(x+3)^2 + (y)^2} = 10 - \sqrt{(x-3)^2 + (y)^2}$$

$$\Rightarrow (x+3)^2 + (y)^2 = 10^2 - 2 \cdot 10 \cdot \sqrt{(x-3)^2 + (y)^2} + (x-3)^2 + (y)^2$$

$$\Rightarrow x^2 + 6x + 9 + y^2 = 100 - 20\sqrt{(x-3)^2 + (y)^2} + x^2 - 6x + 9 + y^2$$

$$\Rightarrow 12x - 100 = -20\sqrt{(x-3)^2 + (y)^2}$$

$$\Rightarrow (3x-25) = -5\sqrt{x^2-6x+9+y^2}$$

$$\Rightarrow (3x-25)^2 = 25(x^2-6x+9+y^2)$$

$$\Rightarrow 9x^2 - 150x + 625 = 25x^2 - 150x + 225 + 25y^2$$

$$\Rightarrow 9x^2 - 150x + 625 = 25x^2 - 150x + 225 + 25y^2$$

$$\Rightarrow 9x^2 - 25x^2 - 25y^2 = 225 - 625$$

$$\Rightarrow -16x^2 - 25y^2 = -400$$

$$\Rightarrow -16x'' - 25y'' = -400$$

$$\Rightarrow 16x'' + 25y'' = 400$$

$$\Rightarrow \frac{x''}{25} + \frac{y''}{16} = 1 \quad [\text{dividing by } 400]$$

(g) Given that,

$$|z_1 + z_2|^2 + |z_1 - z_2|^2 = 2|z_1|^2 + 2|z_2|^2$$

Let, $z_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$

$$|z_1| = \sqrt{x_1^2 + y_1^2} \quad \text{and} \quad |z_2| = \sqrt{x_2^2 + y_2^2}$$

$$\Rightarrow |z_1|^2 = x_1^2 + y_1^2 \quad \text{and} \quad |z_2|^2 = x_2^2 + y_2^2$$

$$\text{L.H.S} = |z_1 + z_2|^2 + |z_1 - z_2|^2$$

$$= |x_1 + iy_1 + x_2 + iy_2|^2 + |x_1 + iy_1 - x_2 - iy_2|^2$$

$$= |(x_1 + x_2) + i(y_1 + y_2)|^2 + |(x_1 - x_2) + i(y_1 - y_2)|^2$$

$$= \{ \sqrt{(x_1 + x_2)^2 + (y_1 + y_2)^2} \}^2 + \{ \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2} \}^2$$

$$= (x_1 + x_2)^2 + (y_1 + y_2)^2 + (x_1 - x_2)^2 + (y_1 - y_2)^2$$

$$= x_1^2 + 2x_1x_2 + x_2^2 + y_1^2 + 2y_1y_2 + y_2^2 + x_1^2 - 2x_1x_2 +$$

$$x_2^2 + y_1^2 - 2y_1y_2 + y_2^2$$

$$\begin{aligned}
 &= x_1^2 + x_2^2 + y_1^2 + y_2^2 + x_1^2 + x_2^2 + y_1^2 + y_2^2 \\
 &= 2x_1^2 + 2x_2^2 + 2y_1^2 + 2y_2^2 + (y_1^2 + y_2^2) + (x_1^2 + x_2^2) \\
 &= 2(x_1^2 + x_2^2 + y_1^2 + y_2^2) + (y_1^2 + y_2^2) + (x_1^2 + x_2^2) \\
 &= 2|z_1|^2 + 2|z_2|^2 + (y_1^2 + y_2^2) + (x_1^2 + x_2^2)
 \end{aligned}$$

$$= \text{R.H.S.} : (y_1^2 + y_2^2 + x_1^2 + x_2^2) +$$

$$= \text{L.H.S.} = \text{R.H.S.} \quad (\text{proved})$$

$$(H) (i) |z - 3 + i| = 4 \Rightarrow y \cdot 2\pi i + x \cdot 2\pi i$$

$$\Rightarrow |x + iy - 3 + i| = 4 \Rightarrow (x-3) + i(y+1)$$

$$\Rightarrow (x-3)^2 + (y+1)^2 = 4^2 \Rightarrow x^2 - 6x + 9 + y^2 + 2y + 1 = 16$$

$$\Rightarrow \sqrt{(x-3)^2 + (y+1)^2} = 4 \Rightarrow x^2 - 6x + 9 + y^2 + 2y + 1 = 16$$

$$\Rightarrow (x-3)^2 + (y+1)^2 = 4^2$$

It is a circle with centre $(-3, 1)$ and radius 4.

$$(ii) |z| = 0$$

$$|z| = |x + iy|$$

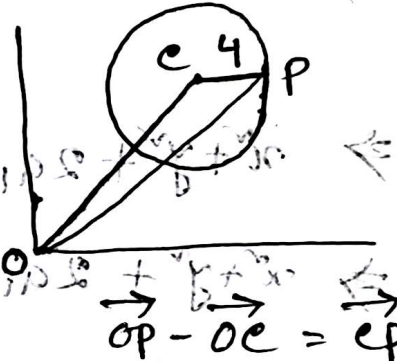
$$\Rightarrow |x + iy| = 0$$

$$\Rightarrow \sqrt{x^2 + y^2} = 0$$

$$\Rightarrow x^2 + y^2 = 0$$

$$x = 0, y = 0$$

The locus is the single point at the origin $(0, 0)$.



$$(j) \quad z\bar{z} + a z + \bar{a}\bar{z} + b = 0$$

$$\Rightarrow (x+iy)(x-iy) + (a_1+ia_2)(x+iy) + (a_1-ia_2)(x-iy) + b = 0$$

$$\Rightarrow x^2 - (iy)^2 + (a_1x + a_1iy + ia_2x + ia_2iy) + (a_1x - ia_1iy - ia_2x + ia_2iy) + b = 0$$

$$\Rightarrow x^2 + y^2 + 2a_1x + 2ia_2y + b = 0 \quad (H)$$

$$\Rightarrow x^2 + y^2 + 2a_1x + 2ia_2y + b = 0 \quad (H)$$

$$\Rightarrow x^2 + y^2 + 2a_1x + 2ia_2y + b = 0 \quad (H)$$

$$\Rightarrow x^2 + y^2 + 2a_1x + 2ia_2y + b = 0 \quad (H)$$

$$\Rightarrow x^2 + y^2 + 2a_1x - 2a_2y + (b + \epsilon) = 0 \quad (H)$$

which is a circular. = $(1+\epsilon) + (\epsilon-x)$

Consider point $(1,0)$ orthocenter of triangle

$$0 = |z| \quad (ii)$$

$$|b+x| = |z|$$

$$0 = |b+x|$$

$$= (b+x)$$

Q. (L) Given that, $(\frac{1}{b} - \frac{1}{b} e^{i\theta}) i + (\frac{1}{b} x e^{-i\theta} - \frac{1}{b} x) =$

$$f(z) = z^3 + 2 \operatorname{Re} \{ z^{\sim} + z \bar{z} + 1 \} + \operatorname{Im} \{ |z|^{\sim} \}$$

We know,

$$z = x + iy$$

$$\bar{z} = x - iy$$

$$\begin{aligned} z^{\sim} &= (x + iy)^{\sim} \\ &= x^{\sim} + 2x \cdot i y^{\sim} + (iy)^{\sim} \\ &= x^{\sim} - y^{\sim} + 2i xy \end{aligned}$$

Conjugate

$$\frac{y^{\sim}}{x^{\sim}} = \frac{y^{\sim}}{x^{\sim}}$$

$$|z|^{\sim} = x^{\sim} + y^{\sim}$$

$$= (x^{\sim})^2$$

Now, $z^3 = (x + iy)^3$

$$= x^3 + 3x^{\sim} \cdot iy + 3x \cdot (iy)^{\sim} + (iy)^3$$

$$= x^3 + 3ix^{\sim}y - 3xy^{\sim} - i y^3 = [i(3x^{\sim}y - y^3)]$$

$$z \bar{z} = (x + iy)(x - iy)$$

$$= x^{\sim} - (iy)^{\sim}$$

$$= x^{\sim} + y^{\sim}$$

$$f(z) = x^3 + 3ix^{\sim}y - 3xy^{\sim} - i y^3 + 2 \operatorname{Re} \{ x^{\sim} - y^{\sim} + 2i xy + (x + iy)(x - iy) + 1 \} + \operatorname{Im} \{ x^{\sim} + y^{\sim} \}$$

$$= (x^3 - 3xy^{\sim}) + i(3x^{\sim}y - y^3) + 2 \operatorname{Re} \{ x^{\sim} - y^{\sim} + 2i xy + x^{\sim} + y^{\sim} + 1 \} + \operatorname{Im} (x^{\sim} + y^{\sim})$$

$$= (x^3 - 3xy^2) + i(3x^2y - y^3) + 2(2x^2 + 1) + 0$$

$$= (x^3 - 3xy^2) + i(3x^2y - y^3) + 4x^2 + 2$$

Cavehy Remain,

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$$

$$\frac{\partial u}{\partial x} = - \frac{\partial v}{\partial y}$$

$$\left. \begin{aligned} \frac{\partial u}{\partial x} &= \frac{\partial v}{\partial y} \\ \frac{\partial u}{\partial x} &= - \frac{\partial v}{\partial y} \end{aligned} \right\} \begin{aligned} \psi(z) &= z \\ \psi(z) &= z^2 \end{aligned}$$

$$u(x, y) =$$

$$x^3 + y^3 - 3xy^2 =$$

$$u = x^3 - 3xy^2 + 4x^2 + 2 \quad \psi(z) = z^2$$

$$v = 3x^2y - y^3 + 8x + 0$$

$$u(x, y) = x^3 - 3xy^2 + 4x^2 + 2$$

$$\frac{\partial u}{\partial x} = x^2 - 3y^2 + 8x + 0$$

$$\frac{\partial v}{\partial y} = 3x^2 - 3y^2$$

$$\frac{\partial u}{\partial x} \neq \frac{\partial v}{\partial y}$$

$$\psi(z) = z^2 + \frac{1}{2}z^4$$

$$\psi(z) = z^2 + \frac{1}{2}z^4 = (x^2 - y^2 + i2xy) + \frac{1}{2}(x^4 - 6x^2y^2 + y^4 + i4x^3y - 4xy^3) = (x^2 - y^2 + \frac{1}{2}x^4 - 3x^2y^2 + \frac{1}{2}y^4) + i(2xy + \frac{2}{3}x^3y - \frac{2}{3}xy^3)$$

$$\frac{\partial u}{\partial y} = -6xy$$

$$-\frac{\partial v}{\partial x} = -6xy$$

$$\therefore \frac{\partial v}{\partial y} = -\frac{\partial v}{\partial x}$$

Not analytic and Don't have derivatives.

Laplace equation:

In two dimension, a second order partial differential equation $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$ is called laplace equation for the function $u = f(x, y)$.

Harmonic function:

Any function which satisfies laplace's equation is called harmonic function.

Example:

If $u = f(x, y)$ be a function, then the harmonic will be $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$.

Conjugate harmonic function:

If $f(z) = u + iv$ be an analytic function, where u and v satisfies laplace's equation.

Then v is called conjugate harmonic function of u , and u is called conjugate harmonic function of v .

Determine conjugate harmonic function of u :

$$dv = \frac{\partial v}{\partial x} dx + \frac{\partial v}{\partial y} dy.$$

Determine conjugate harmonic function of v :

$$du = \frac{\partial u}{\partial x} dx + \frac{\partial u}{\partial y} dy.$$

prob. 1.

$\frac{\partial}{\partial y}$

$\frac{\partial}{\partial x}$

$\frac{\partial}{\partial y}$

harmonic

prob.

Show that the following function is harmonic.

(a) $u = y^3 - 3xy^2$ (b) $u = x^3 - 3xy^2$ (c) $u = 2xy + y^3 - 3xy$

(d) $u = \sin x \cosh y$ (e) $u = \sin x \sinh y$

(f) $u = \cos x \cosh y$.

Determine its harmonic conjugate and construct the corresponding analytic function $f(z)$.

Solution:

(a) Given that

$$u = y^3 - 3xy^2$$

The function u will be harmonic if

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0.$$

$$\therefore \frac{\partial u}{\partial x} = 0 - 3y \cdot 2x = -6xy$$

$$\Rightarrow \frac{\partial^2 u}{\partial x^2} = -6y \cdot 1 = -6y \quad \text{--- (1)}$$

$$\text{and } \frac{\partial u}{\partial y} = 3y^2 - 3x \cdot 1 = 3y^2 - 3x$$

$$\Rightarrow \frac{\partial^2 u}{\partial y^2} = 3 \cdot 2y - 0 = 6y \quad \text{--- (2)}$$

Adding (1) and (2), we get

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = -6y + 6y = 0$$

$$\text{Since } \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

So, the given function u is harmonicLet v be the conjugate harmonic of u .

$$\begin{aligned} \therefore dv &= \frac{\partial v}{\partial x} dx + \frac{\partial v}{\partial y} dy \\ &= -\frac{\partial u}{\partial y} dx + \frac{\partial u}{\partial x} dy \end{aligned} \quad \left| \begin{array}{l} \text{from C-R equation} \\ \frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \text{ and } \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x} \\ \Rightarrow -\frac{\partial u}{\partial y} = \frac{\partial v}{\partial x} \end{array} \right.$$

$$\Rightarrow dv = (3y^2 - 3x) dx + (-6xy) dy \text{ which is}$$

$$M dx + N dy \text{ form that means exact form.}$$

$$\int dv = \int M dx + \int (\text{terms of } N \text{ free from } x) dy + C$$

$$\Rightarrow \int dv = \int (3x^2 - 3y^2) dx + C$$

$$\therefore v = \int 3x^2 dx - \int 3y^2 dx + C$$

$$= 3 \int x^2 dx - 3y^2 \int dx + C$$

$$= 3 \cdot \frac{x^3}{3} - 3y^2 \cdot x + C$$

$$\Rightarrow v = x^3 - 3xy^2 + C$$

$\therefore$ The conjugate harmonic function $v = x^3 - 3xy^2 + C$.

Now,

The corresponding analytic function $f(z)$ is

$$f(z) = u + iv$$

$$\Rightarrow f(z) = y^3 - 3xy^2 + i(x^3 - 3xy^2 + C)$$

(d) Given that

$$u = \sin x \cosh y$$

The function u will be harmonic if

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

$$\therefore \frac{\partial u}{\partial x} = \cos x \cosh y$$

$$\Rightarrow \frac{\partial^2 u}{\partial x^2} = -\sin x \cosh y \quad \text{--- (1)}$$

$$\therefore \frac{\partial u}{\partial y} = \sin x \sinh y$$

$$\Rightarrow \frac{\partial^2 u}{\partial y^2} = \sin x \cosh y \quad \text{--- (2)}$$

$$\left| \begin{array}{l} \frac{d}{dy} (\sinh y) = \cosh y \\ \frac{d}{dy} (\cosh y) = \sinh y \end{array} \right.$$

Adding (1) and (2), we get

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = -\sin x \cosh y + \sin x \cosh y = 0$$

$$\text{Since } \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

So, the given function u is harmonic.

Let v be the conjugate harmonic of u .

$$\therefore dv = \frac{\partial v}{\partial x} dx + \frac{\partial v}{\partial y} dy \quad \left| \begin{array}{l} \text{From C-R equation} \\ \frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \text{ and } \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x} \\ \Rightarrow -\frac{\partial u}{\partial y} = \frac{\partial v}{\partial x} \end{array} \right.$$
$$= -\frac{\partial u}{\partial y} dx + \frac{\partial u}{\partial x} dy$$

$\Rightarrow dv = -\sin x \sinh y dx + \cos x \cosh y dy$ which is exact form that means $M dx + N dy$ form.

$$\int dv = \int -\sin x \sinh y dx +$$

$$\therefore \int dv = \int M dx + \int (\text{terms of } N \text{ free from } x) dy + c$$

$$\Rightarrow \int dv = \int -\sin x \sinh y dx + c$$

$$\therefore v = -\sinh y \int \sin x dx + c$$
$$= -\sinh y (-\cos x) + c$$

$$\Rightarrow v = \cos x \sinh y + c$$

The conjugate harmonic function $v = \cos x \sinh y + c$
now. The corresponding analytic function $f(z)$ is

$$f(z) = u + iv$$

$$\Rightarrow f(z) = \sin x \cosh y + i (\cos x \sinh y + c)$$

Adding (1) and (2), we get

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = -\sin x \cosh y + \sin x \cosh y = 0$$

Since $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$

So, the given function u is harmonic.

Let v be the conjugate harmonic of u .

$$\begin{aligned} \therefore dv &= \frac{\partial v}{\partial x} dx + \frac{\partial v}{\partial y} dy & \left| \begin{array}{l} \text{From C-R equation} \\ \frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \text{ and } \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x} \\ \Rightarrow -\frac{\partial u}{\partial y} = \frac{\partial v}{\partial x} \end{array} \right. \\ &= -\frac{\partial u}{\partial y} dx + \frac{\partial u}{\partial x} dy \end{aligned}$$

$$\Rightarrow dv = -\sin x \sinh y dx + \cos x \cosh y dy \text{ which is exact form that means } M dx + N dy \text{ form.}$$

$$\int dv = \int -\sin x \sinh y dx +$$

$$\therefore \int dv = \int M dx + \int (\text{terms of } N \text{ free from } x) dy + c$$

$$\Rightarrow \int dv = \int -\sin x \sinh y dx + c$$

$$\begin{aligned} \therefore v &= -\sinh y \int \sin x dx + c \\ &= -\sinh y (-\cos x) + c \end{aligned}$$

$$\Rightarrow v = \cos x \sinh y + c$$

The conjugate harmonic function $v = \cos x \sinh y + c$.
Now, The corresponding analytic function $f(z)$ is

$$f(z) = u + iv$$

$$\Rightarrow f(z) = \sin x \cosh y + i(\cos x \sinh y + c)$$